MIL-R-13495F
21 February 1986
SUPERSEDING
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31 January 1979

MILITARY SPECIFICATION

RELINERS, BRAKE SHOE AND CLUTCH RIVETING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of floor mounted brake shoe and clutch reliners which are partially, or completely, powered by electric motors.

1.2 Classification. The reliners shall be of the following types, as specified (see 6.2.1):

Type I - Electric motor and foot-powered.
Type II - Electromechanically powered.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

W-C-596	-	Connector, Electrical, Power, General Specification for.
W-C-596/13	-	Connector, Plug, Electrical, General Purpose, Hospital Grade, Grounding, 2 Pole, 3 Wire, 15 Amperes, 125 Volts, 50/60 Hertz.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-S (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

AMSC N/A

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CC-M-1807 - Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).

MILITARY

MIL-T-152 - Treatment, Moisture- and Fungus-Resistant. Of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.

* 2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1 - General Standards for Industrial Control and Systems.
NEMA MG1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, NW, Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.
NFPA 79 - Electrical Standard for Metalworking Machine Tools and Plastics Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

3.2 Design. The reliners shall be new and of the manufacturer's latest design that shall meet the performance requirements specified in this specification. The basic design shall include a pedestal base with separate stations for drilling and countersinking, and for riveting and deriveting brake shoes and clutch plate linings. A station for grinding brake shoe linings shall also be provided. The stations shall be mounted on the pedestal base as a complete unit. The design shall allow for accessible overhaul and maintenance as well as safe and effective operation. The design shall be such that all parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement without modification, distortion, or damage to the reliner or any reliner component.

* 3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the reliners. In this specification, all measurements, dimensions, sizes, and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in FED-STD-376.

* 3.2.2 Measuring and indicating devices. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and SI systems

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of measurement. When a dual US and SI system is furnished, dials shall have independent zero adjustments and shall be calibrated so that the last dial graduation progresses into, and is continuous with, the first dial graduation as the dial is rotated through the zero position. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the reliners shall be graduated in the same system.

3.2.3 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

* 3.2.4 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

* 3.2.5 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards. Guards shall be removable to provide access to guarded parts. The reliners shall comply with the standards promulgated under OSHA 2206 that are applicable to the reliners themselves. The Type II reliner shall be equipped with a safety feature which shall prevent the reliner from inadvertently repeating while riveting or deriveting. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.4).

* 3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.4 Construction. The reliners shall be constructed of parts which are new, without defects, and free of repairs. The structures shall be capable of withstanding all forces encountered during operation of the machines to their maximum rating and capacity without distortion.

* 3.4.1 Castings and forgings. Castings shall be free from visible blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from visible scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

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* 3.4.3 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when subject to renewal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

* 3.4.4 Lubrication. All bearings (except sealed, permanently lubricated type), mating gears, and other moving parts shall be provided means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. Oil holes, grease fittings, and filler caps shall be located to be accessible. The manufacturer shall include and install all lubricants, fluids, greases, and other compounds in accordance with the best commercial practice of the producing industry (see 6.5). The lubrication systems shall be temporarily capped or stopped to prevent loss of lubricants during shipment. The manufacturer shall include a warning notice in the instruction manual and on the reliners to remove the temporary caps or stoppers before start-up or use of the reliners.

3.4.5 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in soldering operations.

3.5 Performance and product characteristics. The reliner shall perform the operations required to deline and reline brake shoes and clutch plates. The operations shall include deriveting, drilling, countersinking, riveting, and grinding. The grinding, drilling, and countersinking operations shall be electrically powered. The deriveting, riveting, and clinching shall be foot-powered on the Type I, and electromechanically on the Type II reliners.

* 3.5.1 Riveting station. The station, also capable of deriveting, shall basically consist of a "C" type frame of cast iron or welded steel construction. The knockout and clinching mechanism shall be adaptable to receive each of the complete range of standard knockout and clinching holders, as specified herein. Adjustment shall be provided to control the entry of the punch into the anvil bushing to a depth of not less than 0.50 inch. The deriveting punch and riveting clincher shall be in alignment with the knockout anvil and riveting anvil within 0.020 inch. For the Type I machine, the station shall have a throat depth of not less than five inches. For the Type II machine, the station shall have a throat depth of not less than nine inches.

3.5.2 Drilling and countersinking station. This station shall include a rotating spindle, a sliding work support and spindle depth control, and a hole locator.

3.5.2.1 Spindle. The spindle shall be capable of receiving each size drill and countersink specified herein. A positive locking device shall be provided on the spindle to hold the tools in operating position. The spindle shall be mounted on permanently lubricated, sealed ball or roller thrust bearings. The spindle speed shall be not less than 1725 revolutions per minute. The countersink spindle and countersink shall run true to within 0.005 inch total indicator reading (tir).

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3.5.2.2 Work support. The work support shall operate by pressure generated by pressing the brake shoe on the support. The work support shall be rigidly constructed and free operating with no visible deflection, rocking, or binding in the countersink operation. The path of travel down and over the spindle center shall be controlled by at least two guide pins. At least one of the pins shall be spring loaded to return the work support to starting position when pressure is released. An adjustable stop collar on the other pin, or a separate device, shall be available to regulate the depth for countersinking. The work support and the riveting plunger shall be capable of 10,000 complete cycles without damage or deformation.

3.5.2.3 Hole locator. The hole locator, in a fixed or adjustable and suspended position, shall align and indicate the centerline of holes in the workpieces on the work support for countersinking or drilling and countersinking, respectively. The hole locator shall not interfere with the operation of the spindle or work support. The hole locator shall be in alignment with the countersink spindle within 0.010 inch. The locator rod and supporting elements shall not be bent or deformed to meet this alignment.

3.5.3 Grinding station. This station shall include a spindle, drum type abrasive, and an abrasive drum guard.

3.5.3.1 Grinding spindle. The spindle shall be mounted on permanently lubricated sealed bearings. The spindle shall be provided with ball or roller bearings.

3.5.3.2 Roller rests. Not less than four roller rests shall guide and support brake shoes for grinding. The outside roller on each side of the abrasive drum shall be adjustable to suit the applicable radius of the brake drum the shoe is to fit. The roller rests shall be adjustable, as a unit, to regulate pressure of the brake shoe against the abrasive drum.

3.5.4 Tool tray. A metal tray, accessible to the operator, shall be attached to the reliner. The tray shall be capable of holding the standard tools specified herein and an assortment of various sizes of rivets.

3.5.5 Tool box. A tool box, fabricated from sheet steel not less than 0.0598 inch thick, shall be provided. It shall be complete with a hinged lid or door, with a means provided for fastening the lid or door in a closed position. The box shall be capable of holding the handbooks, tools, accessories, and equipment for each reliner. The tool box may be constructed to accommodate the tool tray.

3.5.6 Pedestal. The pedestal and upright column shall be constructed of cast iron or fabricated steel and shall be capable of supporting the entire reliner without any visible deflection or vibration when any of the relining stations are operated. The base shall have not less than three mounting holes and shall be fabricated or machined to receive the column and foot operated treadle.

3.5.7 Foot-power, type I. The rivet and derivet operations shall be operated by foot-power through the use of a foot operated treadle. A means of adjustment shall be provided to control the length of travel of the anvil or punch.

3.5.8 Electromechanical power, type II. The rivet and derivet operating power shall be transmitted from the motor through a clutch and flywheel

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mechanism. The weight of the flywheel shall be not less than 50 pounds. The clutch shall be engaged by a foot operated treadle. The foot treadle, when depressed and released, shall cause the plunger to make one cycle and stop. The foot treadle shall be hinged for lifting to a vertical position when not in use. An automatic compensator shall prevent overloading of the punch capacity. A safety device shall prevent involuntary tripping of the punch while the motor is running.

3.5.9 Rivet body drill sizes. The reliner shall be capable of drilling brake and clutch linings with the following sizes of rivet body drills: 0.125 inch, 0.140625 inch, 0.15625 inch, 0.1875 inch, and 0.250 inch.

3.5.10 Countersinks. The reliner shall be capable of countersinking the above rivet body holes to 0.250 inch, 0.3125 inch, 0.375 inch, and 0.50 inch in countersunk diameters, respectively, to a depth adjustable up to 0.375 inch.

3.5.11 Brake shoe grinder. The reliner brake shoe grinder shall be capable of grinding brake shoes up to, and including, 7.50 inches in width and from 6 to 24 inches in diameter, inclusive. The surface speed of the grinder shall be not less than 1,350 feet per minute.

3.5.12 Riveter. The reliner shall be capable of deriveting, riveting, and clinching tubular or semitubular brass and steel rivets of the following shank diameters 0.125 inch, 0.140625 inch, 0.15625 inch, 0.1875 inch, and 0.250 inch by 1.250 inches long.

3.5.13 Power drive. Power drive to the countersinking station, the grinding station, and the dust collector shall be supplied by single or multiple, direct or indirect drive motor(s). When belts are required, adjustment shall be provided to maintain proper belt tension. One motor may be used to operate the riveter, countersinker, and grinder through belt drive(s).

3.6 Environmental requirements.

3.6.1 Dust collector system. Both the Type I and Type II reliners shall be provided with a dust collector system that shall meet or exceed the asbestos fiber requirements of Subpart Z, Section 1910.1001 of OSHA 2206. Computation formula, for determining compliance to these requirements, shall be in accordance with OSHA Section 1910.1000, Subpart Z. The dust collector system may be either an integral part of the reliner or a separate system. If furnished as a separate system, the system shall be designed and constructed to become a part of the reliner and, when used together, shall meet or exceed the OSHA requirements. No special tools, adapters, or skills shall be required to attach the separate system.

3.7 Electrical equipment. Unless otherwise specified (see 6.2.1), the electrical equipment and its location shall be in accordance with NEMA ICS 1 and NFPA 79, as applicable.

3.7.1 Motor. Unless otherwise specified (see 6.2.1), motors shall be induction, single phase, squirrel cage type meeting the requirements of CC-M-1807. Motors shall have electrical insulation systems conforming to NEMA MG1, class B. The motor shall be equipped with overload protection and ball bearings for continuous duty operation. Frequency and voltage rating shall be for 50/60 hertz, single-phase, 115 volt ±10 percent, alternating current.

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3.7.2 Motor switch. A dustproof motor switch shall be provided which shall be not less than 10 ampere capacity at 115 volts. The OFF/ON positions shall be marked in a plain, visible, and permanent manner. The switch shall be accessible to the operator.

3.7.3 Power supply cord. Unless otherwise specified (see 6.2.1), the reliner motor shall be equipped with not less than 15 feet of three wire, type S or SO, heavy duty electrical cord conforming to NFPA 70. The power supply cord shall be equipped with a plug connector in accordance with W-C-596/13.

3.8 Additional equipment. One set of deriveting, riveting, drilling, countersinking, and grinding tools shall be provided with each reliner. The set shall be complete to cover the performance requirements within the capacity of the reliner, as specified herein. Unless otherwise specified (see 6.2.1), the set of tools shall be the manufacturer's standard equipment.

* 3.9 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.10 Finish. The finish of all parts shall be clean, smooth, and free of burrs, sharp edges, and sharp corners except when required by the original design. Heavy wear surfaces shall be hardened stainless steel or hard chrome plated.

* 3.10.1 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.11 Product identification. The reliners shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.11.1 Caution sign. The manufacturer shall furnish a caution sign with each reliner. Sign specifications and legend shall be in accordance with OSHA 2206, Subpart Z, Section 1910.1001(g)(ii).

3.11.2 Caution plate. A caution plate shall be affixed to the reliner in a location visible to the operator. The caution plate shall contain the legend specified in 3.11.1 in 0.125 inch size letters. The plate shall be metal or mylar. Decalcomania shall not be acceptable.

3.12 Workmanship. Standards of workmanship shall assure that the reliners shall have the stability, strength, durability, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of

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all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. When first article inspection is required, it shall be applied to a preproduction model or initial production item which conforms to the requirements of this specification (see 3.1). First article inspection shall consist of the product examination in 4.5 and all tests in 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

* 4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (e) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Certificate of Conformance (COC) for dust collecter system (see 4.4.2).
(b) Product examination (see 4.5).
(c) Operational test (see 4.6.1).
(d) Alignment test (see 4.6.6).
(e) Packaging inspection (see 4.7).

* 4.4.1 Sampling. Sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test	Inspection Level	AQL
COC for dust collector system	II	1.0
Product examination	II	1.0
Operational test	II	1.0
Alignment test	II	1.0
Packaging inspection	S-4	4.0

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4.4.2 Certificate of Conformance for dust collector system. A COC, supported by inspection and test data to verify conformance of the dust collector system to OSHA 2206, Subpart Z (see 3.6.1), shall be made available to the Government. The contractor shall make the COC available to the Government prior to, or with, the request to perform acceptance inspection by the Government. This is in addition to, and not in lieu of, any rights of the Government under this contract or law. A COC may be used as an element incident to, but shall not be used as the sole basis for, Government acceptance of contract item(s). As a minimum, the COC shall contain the following:

(a) Name of manufacturer and date.
(b) Contract number, National Stock Number, and drawing number (when applicable).
(c) Complete nomenclature of supplies together with lot number or other identification. The quantity in each lot or shipment shall be given.
(d) A statement, as follows, certifying that material meets all requirements of the contract:

The undersigned individually, and as the authorized representative of the contractor, warrants and represents that: All of the information supplied herein is true and accurate; the items covered by this certificate conform to all contract requirements (including but not limited to the drawings and specifications) and this certificate is made for the purpose of inducing payment and with knowledge that the information and certification may be used as a basis for payment.

(e) Signature and title of certifying official.

* 4.5 Product examination. Visually and manually examine the reliners to determine conformance with the requirements of 3.2 through 3.4.5, 3.5.3 through 3.5.8, and 3.7 through 3.12. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the reliners have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.6 Tests.

4.6.1 Operational test. The reliners shall be operated to verify that all hand, foot, or power operated devices and all motors, switches, controls, operating mechanisms, and safety devices operate properly.

4.6.2 Deriveting. Two brake shoes, with linings attached, shall be deriveted and the linings removed. One brake shoe shall have 0.125 inch body diameter rivets and the other shall have 0.250 inch body diameter rivets. The brake shoes shall be examined to assure complete removal of the linings and fall-out of the old rivet (see 3.5.1 and 3.5.12).

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4.6.3 Countersinking. Two new linings of 0.1875 inch basic thickness, clamped in position on the brake shoe, shall be countersunk to suit the rivet sizes. One lining shall be countersunk for a 0.125 inch body diameter rivet and the other for a 0.250 inch diameter rivet. The countersunk linings shall be examined to assure that the sizes conform to the rivet head requirements (see 3.5.10).

4.6.4 Riveting. Linings shall be riveted to two brake shoes, one using 0.125 inch body diameter rivets and the other using 0.250 inch body diameter rivets. The rivets and linings shall be examined to assure that the rivets are not cracked or loose, and that the linings are not loose on the shoes (see 3.5.12).

4.6.5 Grinding. Two sizes of brake shoe and lining assemblies, within the width and diameter range specified in 3.5.11, shall be ground. The grinding station shall be examined during the operation to assure control of stock removal and full contact support of the roller.

4.6.6 Alignment test. Reliners shall be tested to verify conformance with the alignment tolerances specified in 3.5.1, 3.5.2.1, and 3.5.2.3.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of MIL-M-18058 or ASTM D3951-82, as applicable.

5. PACKAGING

* 5.1 Military preservation, packing, and marking. When specified (see 6.2.1), requirements for Level A preservation, Levels A and B packing, and military marking shall be in accordance with MIL-M-18058.

* 5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

6. NOTES

6.1 Intended use. The reliners are intended for use in the relining of brake shoes and clutch plates on general purpose automotive vehicles.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Measuring and indicating device graduation(s) required (see 3.2.2).
(d) Exceptions and additional safety/health requirements when required (see 3.2.5 and 6.4).
(e) Electrical equipment if different (see 3.7).
(f) Motor requirements if different (see 3.7.1).
(g) Power supply cord if different (see 3.7.3).
(h) Additional equipment if different (see 3.8).
(i) Painting and finishing if different (see 3.10.1).

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(j) Product identification if different (see 3.11).
(k) Level of preservation, packing, and marking if different (see 5.1 and 5.2).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

* 6.3.1 Repair part manuals. Repair part identification, in repair part manuals, should include the prime contractor's name, nomenclature, part number, equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items should be identified as shown in MIL-M-7298.

* 6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

* 6.5 Standard lubricants. The reliner manufacturer should specify and use to the maximum extent possible those lubricants (see 3.4.4) that are presently in the Department of Defense supply system. MIL-HDBK-113, MIL-HDBK-267, and MIL-STD-838 may be used as selection guides.

6.6 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:

Army - AL
Navy - YD
Air Force - 99

Preparing activity:

Army - AL

Project No. 4910-0606

Review activities:

Air Force - 82
DLA - CS

User activity:

Navy - MC

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